SCarFFF plotting

Running the plots

Plotting in `SCarFFF` is now handled separately from the calculation, so that figures can be redrawn without recomputing anything. This is done by running:

```
python3 plot.py runs/[run_name] [config/plotting.yaml]
```

The configuration path is optional and defaults to `config/plotting.yaml`, which can be set once and left once you have a good idea of the kinds of plots you want. Alternatively, you might like to create additional plotting config files for different purposes. Every figure requested in the config file is drawn for each saved system, whilst the ones that do not apply (e.g. a structure function mode for a form factor) are skipped. This lets one config file work for both molecular and crystal runs. Figures are saved to each numbered system's `plots/` directory, with the overview at the top and the rest in `full/`, `half/` and `third/`, depending on the sizes requested. LaTeX, dvipng and Ghostscript are needed for the labels.

A plotting configuration

```
file_type: [png, pdf]
width: [full, half, third]
shape: [square, rectangular]
molecules: all
grouping:
  energy_tolerance_eV: 0.01
figures:
  - type: overview
  - type: intensity
    states: [3]
    planes: [xy, xz, yz]
  - type: f_sq_lm
    states: dominant
    modes: dominant
    width: half
  - type: structure_mode
    modes: [[0, 0], [2, 2]]
```

The three settings at the top, `file_type`, `width` and `shape`, are the defaults every figure inherits. Specifying them again inside the figure section of the config overrides them for that specific figure. The `states` option determines which states are plotted for molecules, and can be `all`, `dominant` or a list of transition indices. The `modes` option determines which angular modes are plotted, and can be `dominant` or a list of $[\ell, m]$ pairs. The `energy_tolerance_eV` option sets the energy tolerance for grouping transitions into degenerate groups, which is only used for molecular form factor intensity plots. The `planes` option sets which slices of the form factor are plotted, and can be `xy`, `xz` or `yz`, corresponding to the $q_z = 0$, $q_y = 0$ and $q_x = 0$ planes, respectively.

Available figures

Type	Method	What it shows
`overview`	all	Four panels on one page. Molecular: $z = 0$ intensity slices for the two dominant transitions, the angular mean of the dominant transition, and its angular standard deviation. Crystal: the angular average of the structure function, the three dominant angular modes, and their energy integrals.
`intensity`	molecular	$\|f_s(\vec{q})\|^2$ in the $q_x = 0$, $q_y = 0$ or $q_z = 0$ plane.
`real`, `imag`	molecular	$\mathrm{Re}[f_s(\vec{q})]$ and $\mathrm{Im}[f_s(\vec{q})]$ in the same planes.
`f_sq_lm`	spherical	The intensity spherical harmonic coefficients, $f^2_{s,\ell m}(q)$.
`transition_density`	Fourier	$\Phi^{g\to s}(\vec{r})$ in Å^{-3} in the $x = 0$, $y = 0$ or $z = 0$ plane. Cropped to where the density is, as the FFT box is typically much larger than the molecule.
`angular_mean`	molecular	$\langle\|f_s(q)\|^2\rangle_\Omega$, the angular average of the intensity.
`angular_std`	molecular	$\sigma_\Omega(\|f_s\|^2)$, the angualr standard deviation of the intensity. It measures how anisotropic the response is at each momentum transfer.
`structure_mode`	crystal	One structure function mode, $S_{\ell m}(q, E)$, in $\text{Å}^{-3}\text{eV}^{-1}$.
`structure_modes`	crystal	The dominant anisotropic structure function modes stacked in one figure on a common colour scale: six at full width, two at half, one at third.
`structure_integral`	crystal	Energy integral of the angular averaged structure function, $\int dE\, \langle S(\vec{q}, E)\rangle_\Omega$, in Å^{-3}.
`structure_mode_integrals`	crystal	The same energy integral for the anisotropic modes of the structure function, $\int dE\, S_{\ell m}(q, E)$.
`structure_sum_rule`	crystal	The residual, $\int dE\, [S_{\ell m} - S^{(0)}_{\ell m}]$, with $S^{(0)}_{\ell m}$ the angular modes of the incoherent structure function, which neglects intermolecular interactions. This should look like noise, and so is primarily for verification. Needs `crystal.incoherent_reference` set when the job is run.

Momentum axes are in keV and energy axes in eV. Form factors, their intensities an the coefficients are dimensionless.

Figure sizes

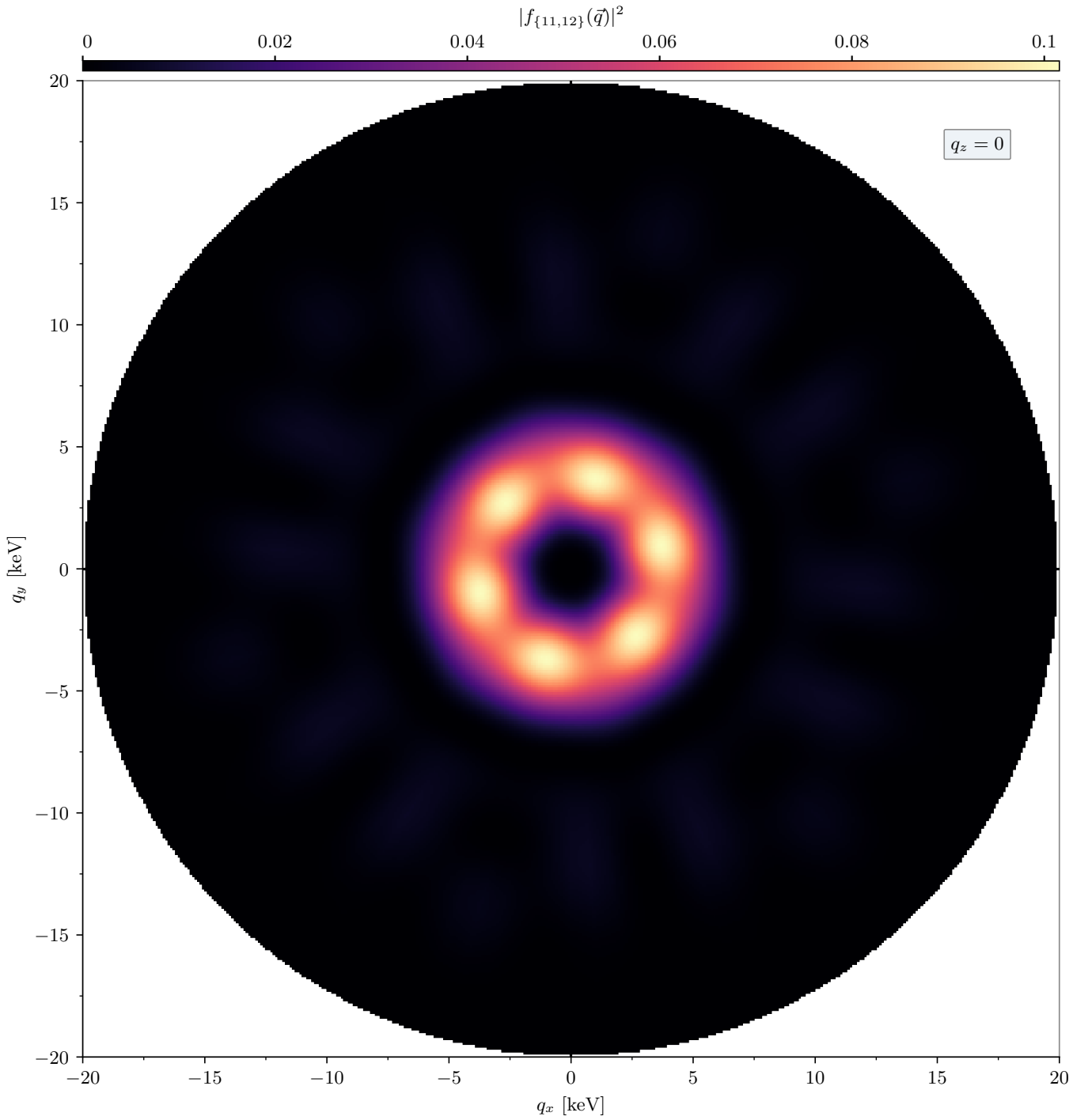
Figures can be drawn at three widths: `full`, `half` and `third`. These are designed to sit on their own, or two/three abreast at the top of a page, with the font sizes set to match the figure caption. Change `text_width_pt` and `caption_pt` under `style` to match the document they are going into if necessary.

Figures can also be drawn square or rectangular. Form factors and third width figuresare always square. Square plots are designed to sit nicely next to form factors on the page, whilst rectangular plots save space for e.g. line plots or structure functions.

Gallery

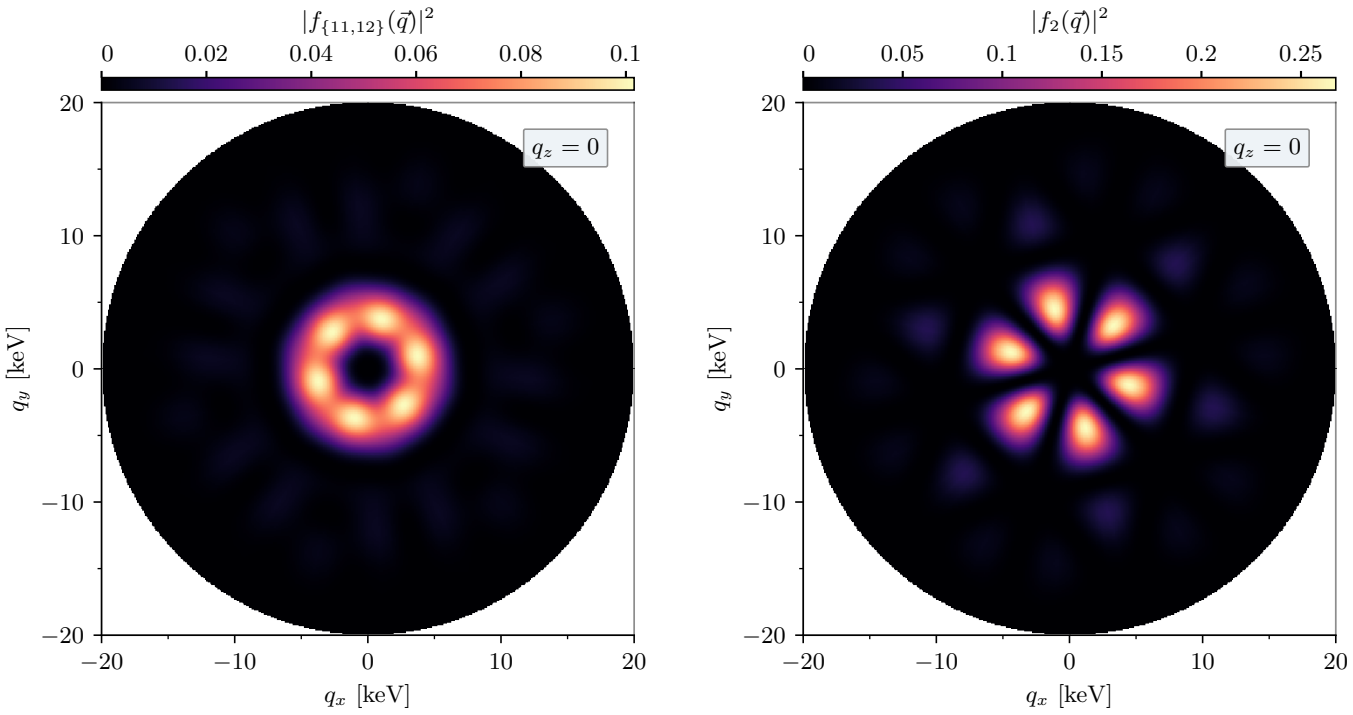
Here we show every kind of figure, and size that can be produced. The molecular figures below are for benzene, computed using the 6-31g* basis set and the B3LYP functional, and on grids of around 100 q points along each axis. The crystal figures are for solid benzene, computed using the def2-SVP basis set and ωB97X-D4 functional, on the same 100 point q grid, and an energy with 10 meV spacing. Where appropriate, all plots use $\ell_{\max} = 24$.

The three widths



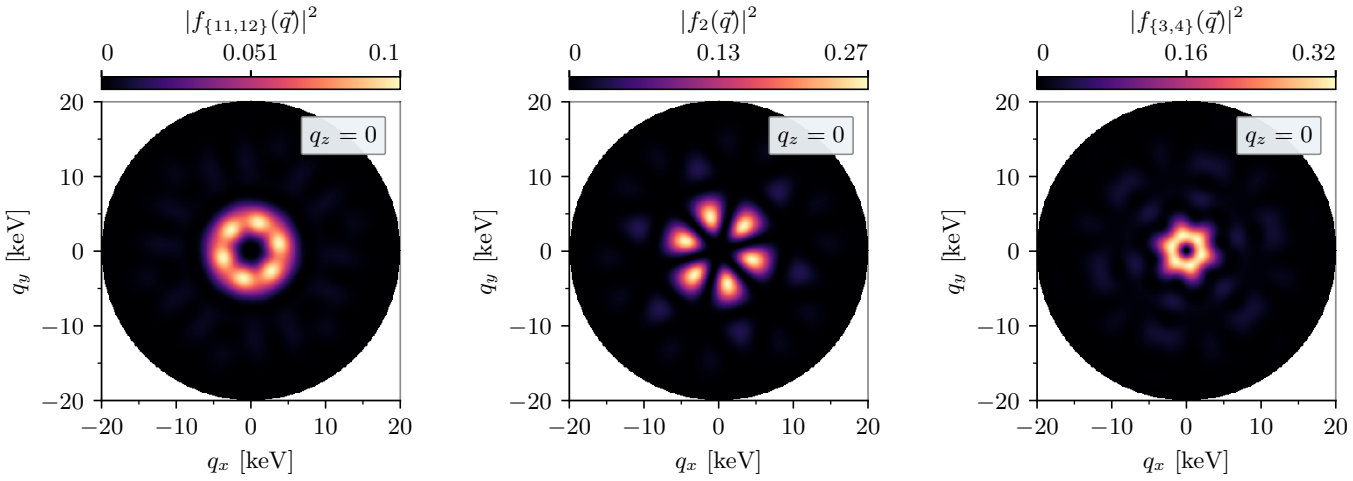
(a) `full`, 510 pt

Figure 1: One full-width figure spans the text width.



(a) `half`, 246 pt **(b)** `half`, 246 pt

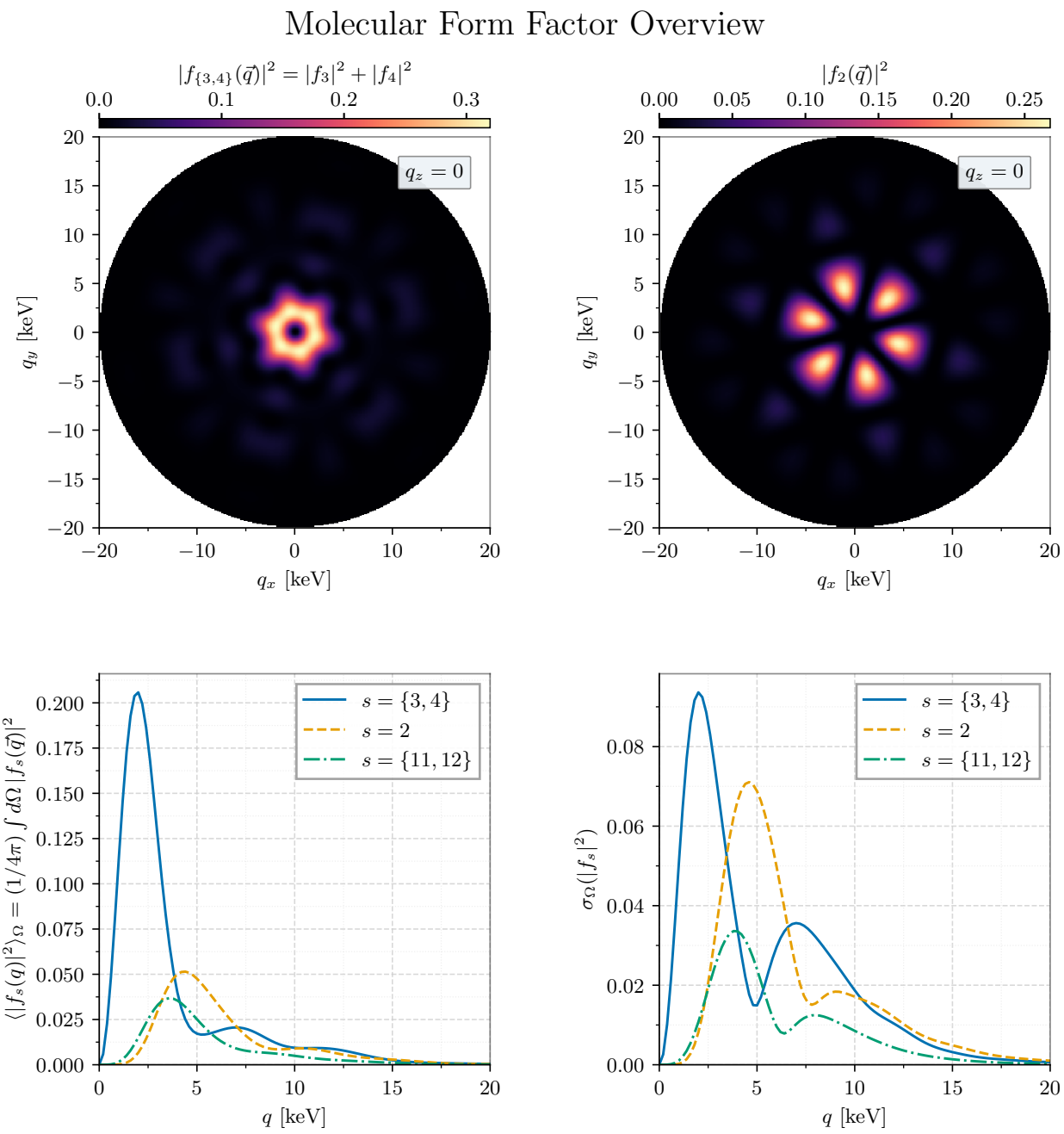
Figure 2: Two half-width figures sit side by side with an small gap between them.



(a) `third`, 158 pt **(b)** `third`, 158 pt **(c)** `third`, 158 pt

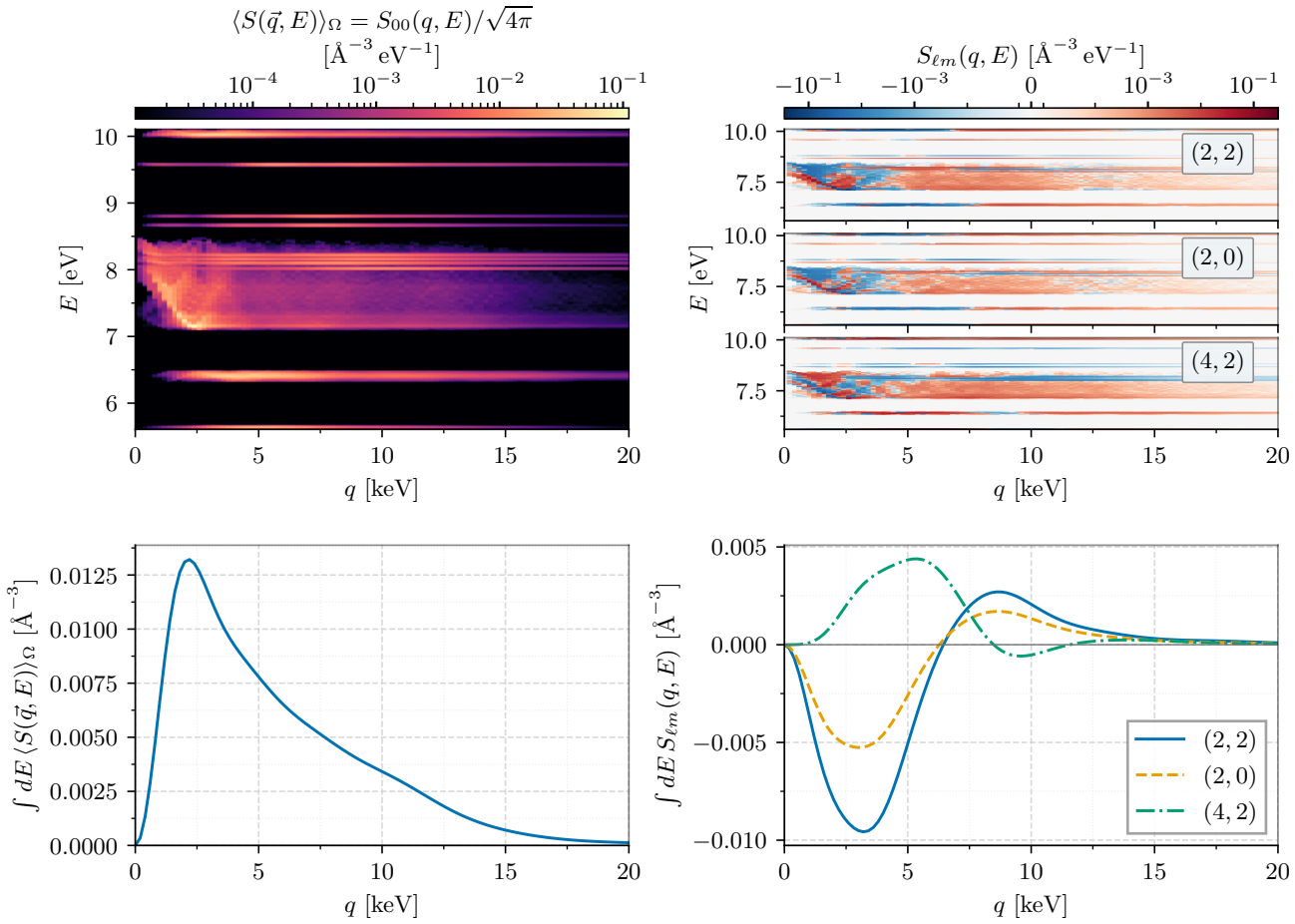
Figure 3: Three third-width figures fill the same width.

One of each figure



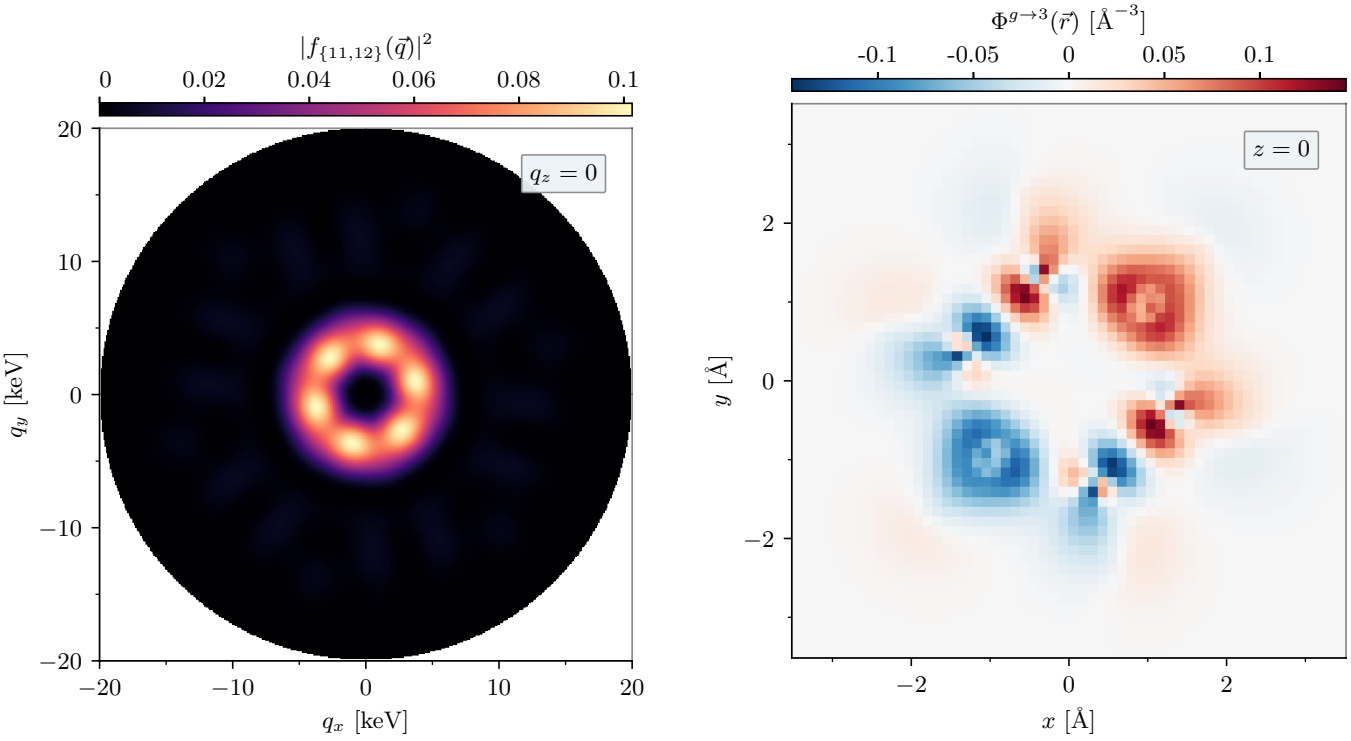
(a) `overview` for a molecule

Figure 4: The molecular overview: two intensity planes above, and the angular mean and standard deviation for the dominant transition below.



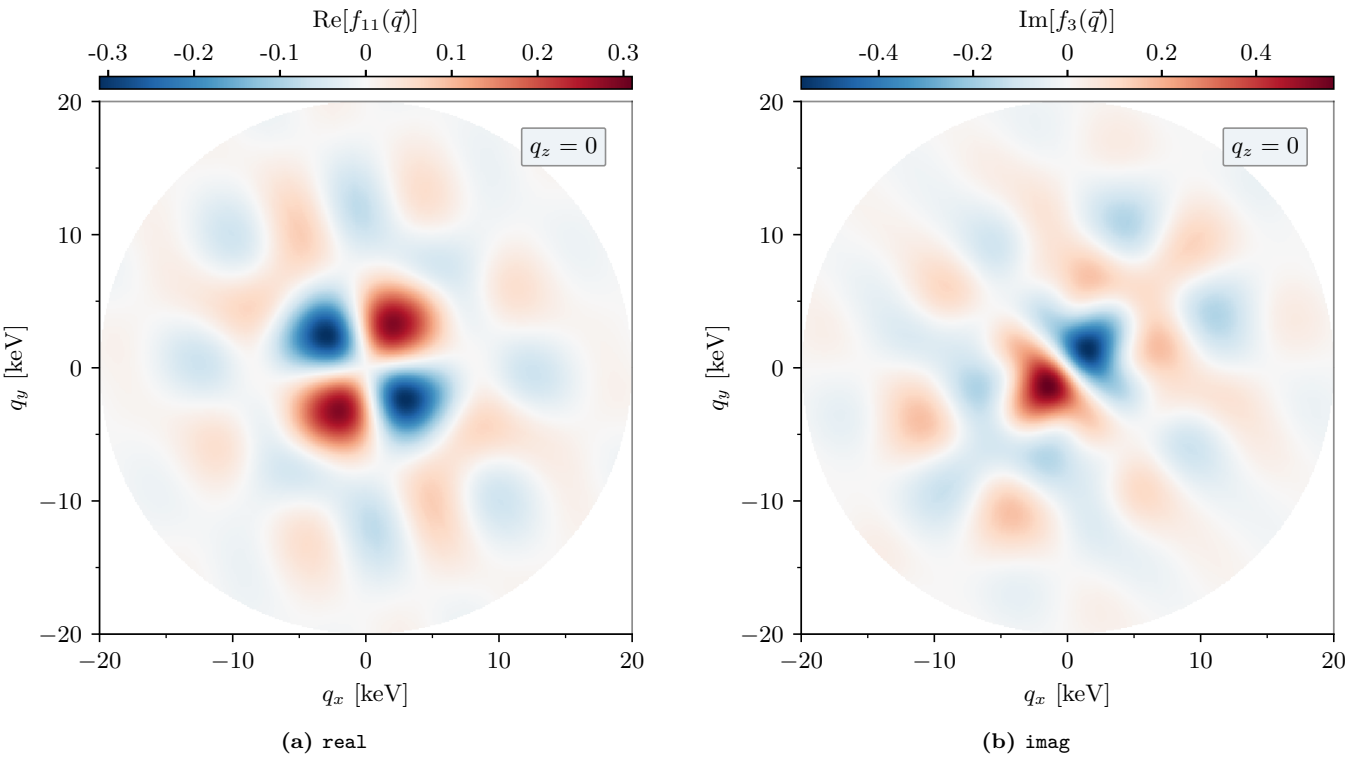
(a) `overview` for a crystal

Figure 5: The crystal overview: the angular average of the structure function and its three dominant angular modes above, their energy integrals below.



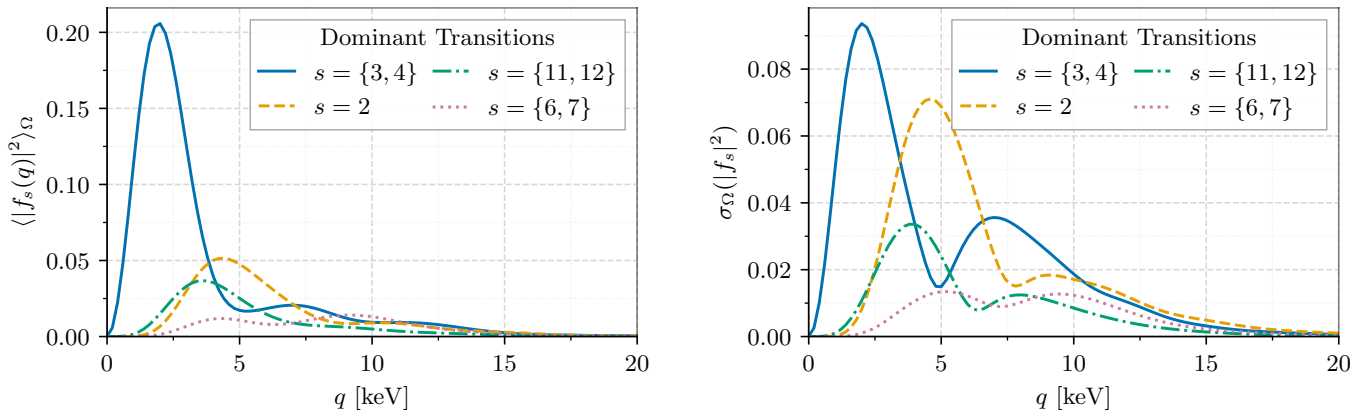
(a) `intensity` **(b)** `transition_density`

Figure 6: A form factor intensity, and a transition density.



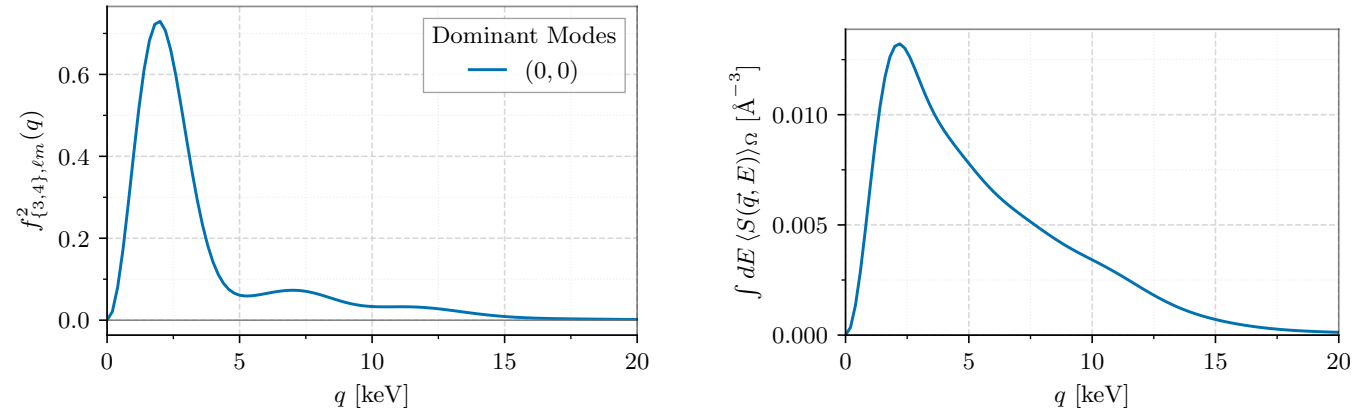
(a) `real` **(b)** `imag`

Figure 7: Real and imaginary parts of two form factors.



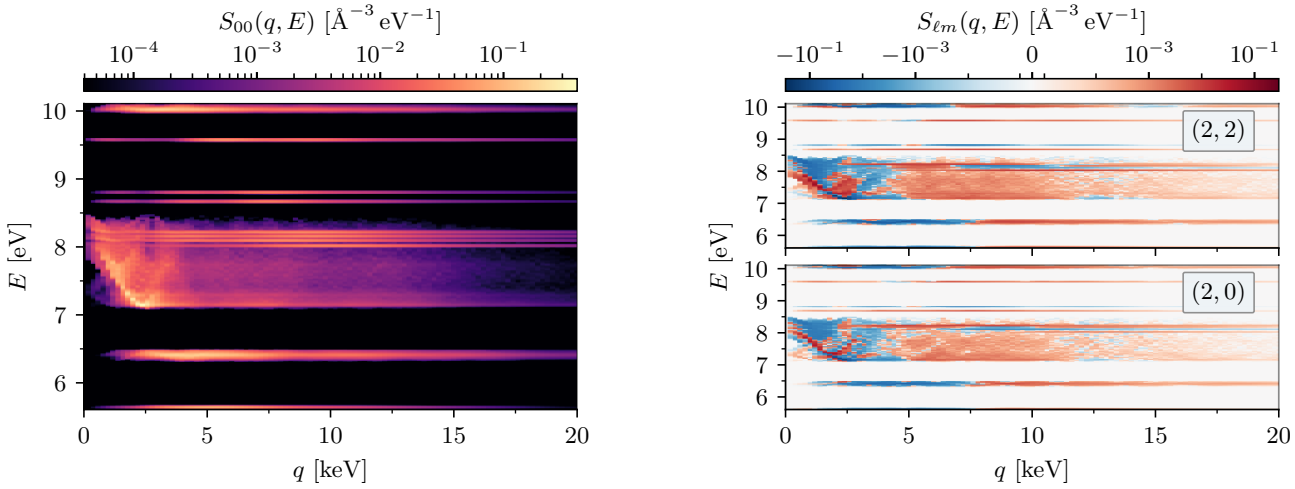
(a) `angular_mean` **(b)** `angular_std`

Figure 8: The angular means and standard deviations the dominant transitions.



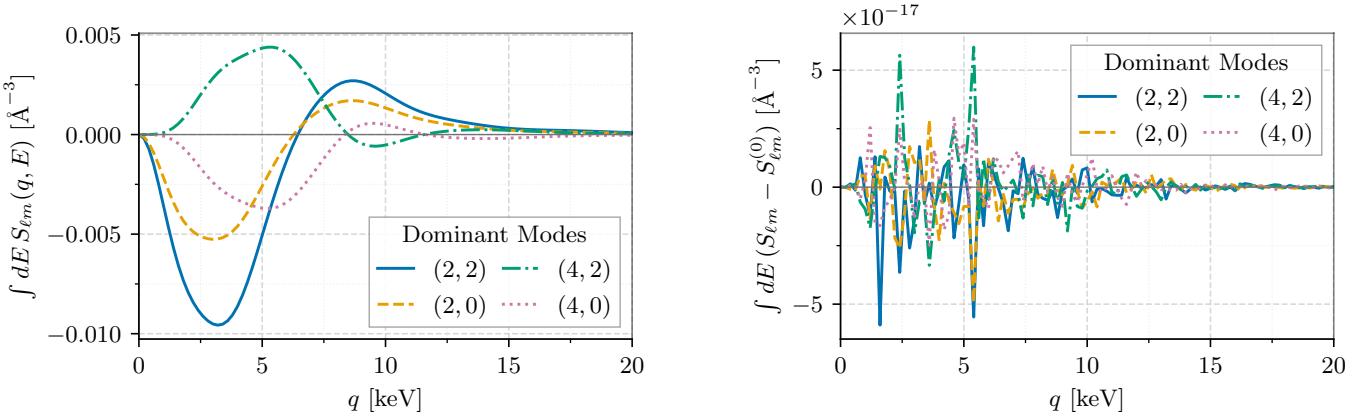
(a) `f_sq_lm` **(b)** `structure_integral`

Figure 9: The intensity angular coefficients, and the energy-integrated angular average of the structure function for the crystal.



(a) `structure_mode` **(b)** `structure_modes`

Figure 10: The isotropic mode of the structure function, and the dominant anisotropic modes.



(a) `structure_mode_integrals` **(b)** `structure_sum_rule`

Figure 11: The energy integral of the dominant anisotropic modes of the structure function, and the sum rule residual.